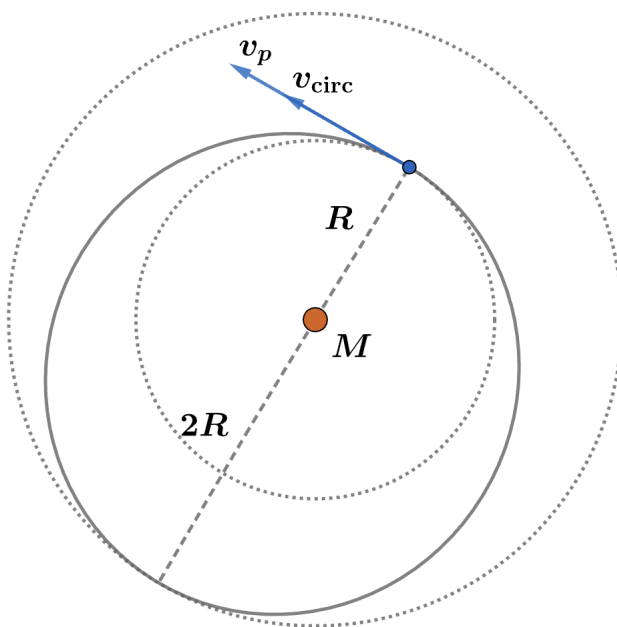


2024 F=ma Exam: Problem 18

Kevin S. Huang



Recall the velocity for a circular orbit is given by

$$v_{\text{circ}} = \sqrt{\frac{GM}{R}}$$

Recall from the elliptical orbit equations that the velocity at the perigee is given by

$$v_p = \frac{a+c}{b} \sqrt{\frac{GM}{a}} = \sqrt{\frac{r_a}{r_p}} \sqrt{\frac{2GM}{r_a+r_p}} = \sqrt{\frac{2R}{R}} \sqrt{\frac{2GM}{2R+R}} = \sqrt{\frac{4GM}{3R}}$$

Then

$$\Delta v = v_p - v_{\text{circ}} = \left(\frac{2}{\sqrt{3}} - 1 \right) \sqrt{\frac{GM}{R}} = 0.15 \sqrt{\frac{GM}{R}}$$

so the answer is B.